

DS-270702 · DATA SCIENCE PROGRAMMING · HOMEWORK 4

PM2.5 in Northern Thailand

From raw data to a recommendation

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Three things can be done about the season

This deck builds the first box out of its own data. The other two rest on published research and on regulation that already exists, and the table says which is which.



Stage	What it means	Where it is in this deck	What the claim rests on
Predict	Know tomorrow before tomorrow	The model, and the first two recommendations	My own test set, 454 days
Prevent	Open rooms and stop bad burning being scored as good	The second and third recommendations	My own data, plus the standard already in force
Recovery	Pay households the season already cost	The fourth recommendation	Published research and the national relief schedule

All three wait on one thing: whether measured station data can be obtained. That is recommendation 1.

THE QUESTION

What I set out to answer

Can tomorrow's PM2.5 in Chiang Mai be predicted well enough to be worth warning people with, and if it can, what should change as a result?

Two reasons this question is worth a term project, and not just curiosity.

Why ask it	What is already true today
The action is already funded and already mandated	37.5 $\mu\text{g}/\text{m}^3$ obliges agencies to hand out masks and open dust-free rooms. A forecast is worth exactly one day of warning on a decision that has already been made
The scoreboard may be measuring the weather	2026 had the longest burning ban on record and the worst fire season on record. When effort and the headline number move in opposite directions, the number needs checking

One command runs everything

Nothing in this deck was produced by hand. Every number comes back if you run one line.

File	What it does
src/config.py	every path, URL and constant, in one place
src/fetch_data.py	downloads, saves the raw response untouched
src/prepare_data.py	clean, join, build features
src/checks.py	the six checkpoints
src/analyse.py	the figures
src/model.py	baseline first, then models, then scoring
run_all.py	runs all of the above in order

```
$ python run_all.py
```

Fetch, prepare, check, draw, model.
Each step wrapped, so one network failure does not stop the rest.

49

raw API responses,
saved before parsing

128,352

hourly rows, from
each of three sources

Four places the code would run fine and answer wrong

Each row is a trap that raises no error. The right column is what the code does instead.

The trap	What the code does
A day with six hours of data averages just as happily as a day with twenty-four	Keep a day only if 18 of 24 hours are present. Otherwise drop it, never average it
The average of 350° and 10° of wind direction is 180°, the exact opposite direction	Average directions as unit vectors, not as numbers
Air4Thai writes a missing reading as the text -1, which averages like a real value	Mask anything at or below -1 before any arithmetic touches it
A feature nobody could know yesterday makes the score look excellent	An allow-list guard stops the run with an error. It covers the weather columns. The forecast columns are a gap I state in the limitations

The data is not measured. It is a model.

38 columns are exactly 0.0000% missing across three and a half years. No instrument is that perfect.

0.0000%

missing, 38 columns,
3.5 years of hourly data

Real monitors lose data to calibration, power cuts and maintenance. Air4Thai writes those gaps as -1.

A complete series is the signature of model output. The source is Copernicus CAMS, which produces a value for every cell and every hour whether or not anything was measured there.

Three things follow from that, and they run through everything after this slide.

Errors are model bias, not random noise	So they do not average away by collecting more days
A model trained on this learns to predict CAMS	So it must be retrained on measured data before anyone relies on it
Every exceedance count here is CAMS's count	So the next checkpoint asks how far off CAMS is

So I measured how wrong it is

70 paired readings against real Air4Thai instruments. CAMS reads low at every single station.

-5.38

$\mu\text{g}/\text{m}^3$ average gap,
CAMS minus instrument

14 / 14

stations where CAMS read
low, with no exception

70

paired readings, collected
over several runs

What it means for my own numbers	Days CAMS reports at 33 to 37 may really have gone over 37.5. My exceedance counts are more likely under-counts than over-counts
Limit I state myself, 1	Measured at 6.9 to 18.4 $\mu\text{g}/\text{m}^3$ in the rainy season. Whether the same gap holds up near 37.5 is not established
Limit I state myself, 2	The gap is uneven, -0.7 in Chiang Mai city and -9.1 at Mae Moh, so the province ranking may partly be uneven model error

The check that passed, and should not have

Two points 12 km apart inside Chiang Mai returned exactly the same numbers for all 168 hours

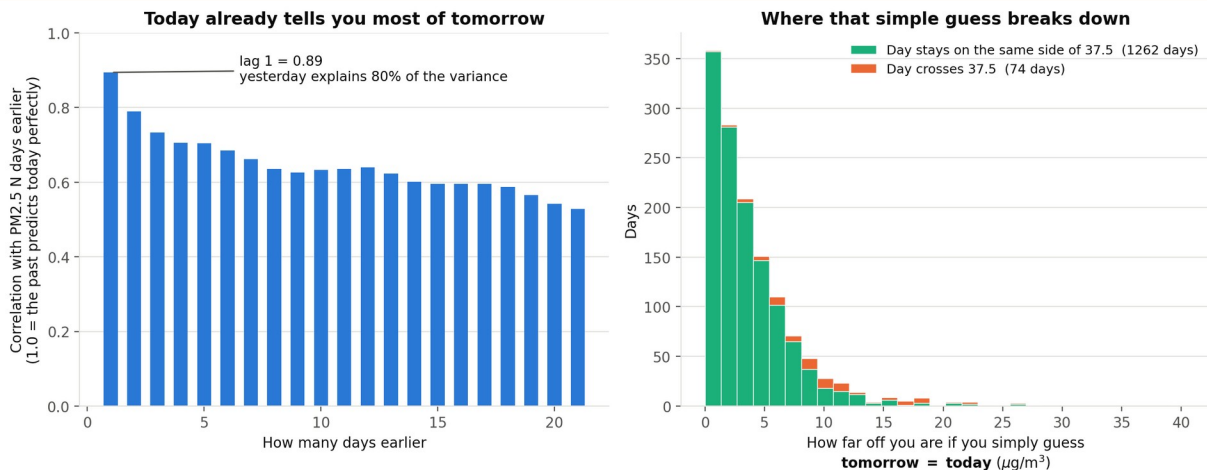
Mueang 18.7883, 98.9853 → grid 18.800 / 99.000
Hang Dong 18.6883, 98.9214 → grid 18.700 / 98.900

**largest difference over
the week: 0.0 µg/m³**

My first version passed	It compared the coordinates the API reported. Six requests, six different coordinates, so it said PASS
What actually causes it	CAMS resolves about 45 km, so two nearby requests are read from the same square. The reported coordinate is cosmetic
How widespread	1 of 15 pairs inside the province identical, 14 different. Only comparing the values finds it
What I did about it	Dropped district-level analysis. Compare the four provincial capitals, checked to be genuinely different

Why build a model at all?

Yesterday alone already explains about 80% of today. So a model has to earn its place somewhere specific.



What kind of day	Share of days	How far off "tomorrow = today" is
Days the air stays on the same side of 37.5	94.5%	3.59 $\mu\text{g}/\text{m}^3$
Days the air crosses 37.5	5.5%	10.60 $\mu\text{g}/\text{m}^3$, three times worse

My model loses on average, and that is the finding

The numbers are the average size of the miss, in $\mu\text{g}/\text{m}^3$. Lower is better. Read the last column.

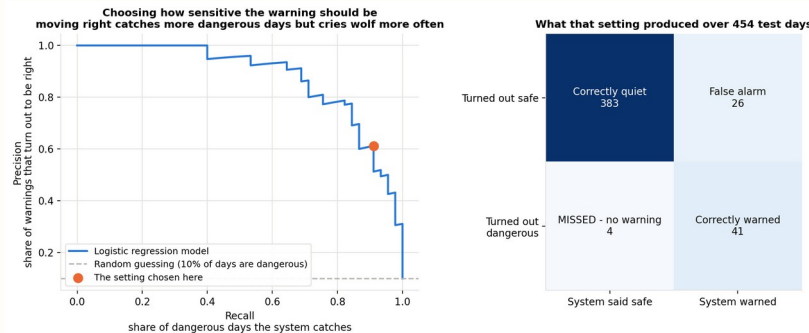
	Ordinary days	Burning season	Days the air changes
Guessing "tomorrow = today"	4.118	6.846	14.079
Ridge regression	4.303	6.529	11.791
Gradient boosting	3.946	6.424	12.098
On ordinary days it barely wins	3.95 against 3.98, which is smaller than the 5.38 gap the data itself carries. Not worth claiming		
On the days that matter it wins clearly	14.1 down to 11.8, a 16% cut, and both models manage it		

The useful output is a warning that tomorrow changes, not tomorrow's number.

That is what the next slide predicts, and what the first recommendation asks the province to publish.

Where to draw the line is a policy choice, not a default

The model returns a probability. Someone has to decide how sure is sure enough to warn.



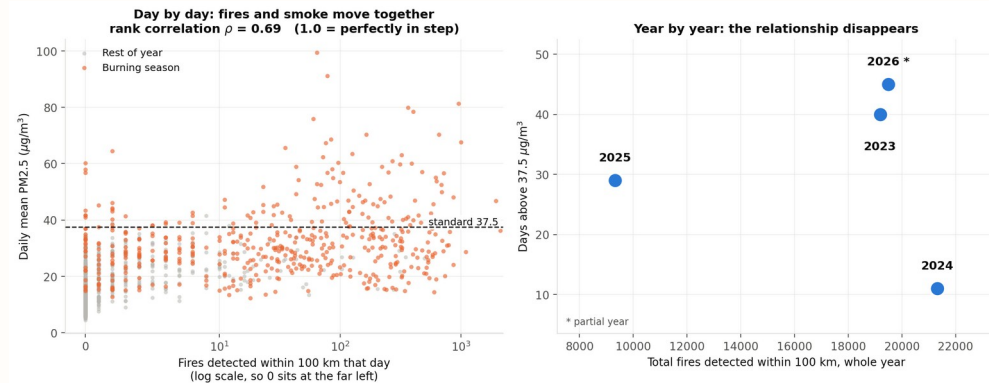
Approach	Dangerous days missed	False alarms	Caught
Say "safe" every day	45	0	0%
Guessing "tomorrow = today"	8	8	82%
Where I set the line	4	26	91%
Tuned to never miss one	0	99	100%

A false alarm costs one day of opening a room and handing out masks. A miss costs a day of people breathing it unprotected. Catching all 45 would mean warning every four and a half days, which is how a warning system teaches people to ignore it.

THE FINDING

A bad-day count is not a burning score

Both are true at once: day to day fires and smoke track closely, year to year they do not.



Year	Fire detections within 100 km	Days over the standard	
2023	19,190	40	
2024	21,312	11	most fires, fewest bad days
2025	9,329	29	fewest fires, nearly 3x 2024's bad days
2026	19,491	45	241 days only

RECOMMENDATIONS

Four, each addressed to a body that can act

Same four boxes every time: who does it, what they do, what it should produce, and what limits it.

PIC	What to do	Expected outcome	Challenge
CMU CCDC and the working group	Obtain the measured district-level station data that already exists	Removes the 5.38 bias and the 45 km limit at once	The data exists. The obstacle is an approval nobody has asked for
Health Center 1 Chiang Mai	Publish each evening the probability that tomorrow goes over 37.5, with the miss rate	Days arriving unwarned fall from 8 to 4 in 454	Predicts CAMS, not the air, until the data above arrives
Provincial PM2.5 working group	Report a burning number and an exposure number side by side	A bad-burning year can be told from a bad-weather year	Does not say what the yearly variation does track
Disaster Prevention and Mitigation	Add an air-pollution category to the national relief schedule	Relief triggers on exposure, not on catastrophe	Only the exposure counts come from this work

The first one is not a caveat moved to the front. Three of the six checkpoints point at it, and the other three recommendations are waiting on it.

HONESTY

What this analysis does not support

I would rather say these myself than be asked.

This does not support	Why
The true number of exceedance days	Every count is CAMS's, and CAMS reads low
Any trend	Four years, one of them incomplete
Seasonal forecasting	Four burning seasons cannot validate a model of next year
City versus countryside in Chiang Mai	The grid is 45 km, wider than the difference
Causation	Fires near a place do not prove those fires made its smoke
Whether the burning ban worked	That needs enforcement records and a comparison group

The one thing that would fix the most: five years of **MEASURED** station data from the **CMU DustBoy** network. It removes the model-versus-instrument problem and the resolution problem at the same time.

What I would want you to remember

The data turned out to be a model, so I measured how wrong it was

-5.38, low at 14 of 14 stations

Two points 12 km apart were the same data

and the API said otherwise

The model loses on average and wins where it counts

14.1 to 11.8 on the days that change

The number the province is judged by tracks the weather

most fires, fewest bad days, same year

Thank you. Repository: github.com/torthanantasetth/dsc-cmu-pm2.5-assignment